

Study Design

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1. Overview

This document describes the prospective observational study design for evaluating the clinical asses

2. Study Population

2.1 Eligibility Criteria

****Inclusion:****

- Incident surgical referrals to Site A or Site B
- Referral date between 2025-08-01 and 2026-07-31
- Any surgical service

****Exclusion:****

- Referral date before study window
- Missing referral timestamp

2.2 Population Definition

2.3 Study Timeline

3. Causal Framework

3.1 Causal Diagrams

****Claim 1 (Cancellation):****

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U -> S -> T -> R -> B -> Y\_cancel

U -> Y\_cancel

calendar -> T, P, Y\_cancel

N -> I -> B, Y\_cancel

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****Claim 2 (Readiness):****

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U -> S -> T -> R -> B -> Y\_ready

U -> Y\_ready

calendar -> T, P, Y\_ready

reached\_readiness <- Y\_ready, U

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****Claim 3 (Agreement):****

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U -> S -> T -> R -> clinician_action -> agreement
model_family -> R, LLM_judge

...

Where:

- U = baseline case mix
- S = service selection
- T = own-service switch
- P = scheduling policy
- N = neighboring services
- I = interference
- R = assessment/recommendation
- B = clinician behavior
- Y = outcome

3.2 Exposure Assignment

****Primary Exposure:**** Service-level display status at referral time

****Unit of Analysis:**** Episode (surgical case)

****Unit of Inference:**** Baseline clinician-coverage connected component

4. Estimands

4.1 Claim 1 Estimand

4.2 Claim 2 Estimand

4.3 Claim 3 Estimand

5. Analysis Methods

5.1 Approach A: Overlap-Weighted Estimators

****C1:**** Overlap-weighted, doubly robust difference-in-differences

****C2:**** Discrete-time marginal structural model with censoring weights

****C3:**** Two-phase inverse-probability weighting

5.2 Approach B: Synthetic Controls

****C1 & C2:**** Augmented synthetic control on service-week outcomes

****C3:**** Pattern-mixture identification region

5.3 Small-Cluster Inference

- Independent unit: Baseline clinician-coverage component
- Methods: CR2 cluster-robust covariance, wild bootstrap-t
- Minimum: 6 components, 2 per arm, 4+ df

6. Covariate Specification

6.1 Adjustment Sets

6.2 Baseline Variables

- Site
- Canonical service code
- Referral week
- Pre-switch service volume/trend
- Patient age, sex
- Ward ID, clinician coverage

7. Missing Data Handling

7.1 Primary Assumption

Missing at random (MAR) conditional on:

- Baseline site/service/case mix
- Calendar time
- Source
- Pre-time-zero history

7.2 Sensitivity Analyses

7.3 Administrative Censoring

- Capped at 2026-07-31
- Addressed with censoring weights
- Late ingests may revise pre-cutoff event time only

8. Multiple Testing Control

8.1 Test Families

8.2 Decision Rule

Effect is "absent" only when both approaches and all missingness analyses place the bias-aware inter

- C1: [-0.02, +0.02] risk difference
- C2: [-3, +3] days
- C3: [-0.05, +0.05] chance-excess

9. Subgroups and Sensitivities

9.1 Subgroups

- Site (A, B)
- Service
- Sex
- Age (<40, 40-64, >=65)

9.2 Sensitivity Analyses

- 30/60/90-day horizons
- Event log vs. snapshot
- Alternate service-code reconciliation
- Policy timing ?2, ?4 weeks
- Weight trimming 1/99, 5/95
- Alternate clustering by ward
- Leave-one-service-out

10. Limitations and Known Departures

1. Services self-selected (best performers)
2. Only Site A selected services switched
3. Post period is three months only
4. Scheduling policy change co-occurred
5. Clinicians unblinded, share wards
6. Treatment can spill across units
7. Outcomes/reviews non-randomly missing
8. No label adjudication
9. LLM judge shares model family
10. Small cluster count

*This design was frozen before outcome analysis. See analysis-plan-v1.yaml for the machine-readable